

PRODUCT REQUIREMENTS DOCUMENT (PRD)

HRM Application – Phase 1 (Core Modules)

Product Overview

Purpose

The HRM application will digitize and streamline three critical HR processes:

- Leave Management
- Staff Loan Management
- Staff Appraisal Management

The system will enforce HR policies, provide multi-level approval workflows, maintain audit trails, and generate performance and financial insights.

Objectives

1. Eliminate manual and email-based HR processes and replace the existing seamless HRM system.
2. Enforce policy compliance through automated validation.
3. Improve approval transparency and accountability.
4. Provide structured reporting and decision support.
5. Maintain audit logs for governance and executive oversight.

Stakeholders

- Employees
- Line Managers
- HR Department
- Executive Director (ED)
- Managing Director (MD)
- Payroll/Finance

User Roles & Access Scope

Employee

- Submit leave requests
- Submit loan applications
- Complete self-appraisal
- View leave balance
- View loan status & repayment schedule
- View appraisal results

Line Manager

- Approve/reject leave requests
- Complete appraisal evaluations
- View team leave calendar
- View team appraisal data

HR

- Review and approve leave

- Modify or cancel leave
- Review and approve loans
- Confirm loan liquidation
- Create and manage appraisal cycles
- Calibrate appraisal scores
- Generate reports

Executive Director (ED)

- Final leave approval
- Loan approval stage

Managing Director (MD)

- Final loan approval

5. Module Requirements

MODULE 1: LEAVE MANAGEMENT

Leave Types

- Annual Leave
- Sick Leave
- Casual Leave
- Maternity Leave

Leave Policy Rules

- Leave allocated annually upfront
- No carry-forward
- No half-day leave
- Weekends excluded from leave calculation
- Public holidays excluded from leave calculation
- Unused leave forfeited upon resignation
- Emergency leave requests allowed

Leave Workflow

Submission

- Employee selects leave type
- Selects date range
- System auto-calculates working days
- Requests for Leave allowance
- System validates available balance

Approval Chain

1. Line Manager
2. HR
3. Executive Director (ED)

Decision Handling

- Rejection at any stage ends process
- Approval at final stage deducts leave balance
- Approved leave auto-updates department calendar

Modification & Cancellation

- Only HR can modify or cancel approved leave.
- The system recalculates balance upon modification.

Status States

- Draft
- Pending Manager
- Pending HR
- Pending ED
- Approved
- Rejected
- Cancelled

Reporting Requirements

- Department leave calendar
- Individual leave balance dashboard
- Leave utilization summary

MODULE 2: STAFF LOAN MANAGEMENT

Loan Types

1. Personal Loan
2. Compassionate Loan

Both are interest-free.

Eligibility Criteria (System-Enforced)

Employee must:

- Be confirmed staff
- Have above-average last appraisal rating
- Have no active loan
- Not be financially indebted to company
- Loan amount $\leq 33.3\%$ of annual net salary
- Repayment period ≤ 12 months
- Monthly installment $\leq 33.3\%$ of monthly take-home pay

If validation fails, application cannot proceed.

Loan Workflow

Submission

- Employee selects loan type
- Specifies amount and repayment tenure
- System calculates installment
- System validates eligibility

Approval Chain

1. HR
2. Executive Director (ED)
3. Managing Director (MD)

Final approval required from MD.

Repayment Structure

- Fixed monthly installment

- Automatically deducted from payroll
- System tracks outstanding balance
- Loan automatically closes when balance reaches zero

Early Liquidation

- Employee may repay early
- HR confirms payment receipt
- System marks loan as liquidated
- Payroll deductions stop

Resignation Handling

- Outstanding balance deducted from final entitlements
- Loan marked closed

Status States

- Draft
- Pending HR
- Pending ED
- Pending MD
- Approved
- Rejected
- Active
- Closed

- Liquidated

Reporting Requirements

- Outstanding loan dashboard
- Monthly deduction summary
- Individual loan ledger
- Loan repayment tracking

MODULE 3: STAFF APPRAISAL MANAGEMENT

Appraisal Types

1. 90-Day Performance Check
2. 6-Month Confirmation Appraisal
3. Bi-Annual July Appraisal
4. Bi-Annual December Appraisal

Each appraisal type must be configurable as a separate cycles

Appraisal Model

- Hybrid (KPI + Competency-based)
- Quantitative scoring
- Qualitative feedback
- Likert scale used
- Weighted scoring system

Appraisal Workflow

Cycle Creation

- HR creates cycle
- Defines weight configuration
- Sets deadline
- Selects employees

Evaluation Flow

1. Employee completes self-appraisal
2. Manager completes evaluation
3. System calculates weighted score
4. HR calibration/normalization
5. Final score confirmed
6. Results visible to employee

Editing Rules

- HR can edit submissions after deadline
- All edits must be logged

Outcome Linkage

Appraisal results may influence:

- Salary increments
- Promotions

- Bonuses
- Training needs analysis

Status States

- Not Started
- Self Submitted
- Pending Manager Review
- Pending HR Calibration
- Finalized
- Reopened (HR only)

Reporting Requirements

- Top performers report
- Low performers report
- Full performance report
- Appraisal outcome summary
- Training needs analysis report
- Departmental comparison
- Performance distribution analysis

Cross-Module Functional Requirements

Audit Trail

All modules must:

- Log approvals and rejections
- Record timestamps
- Record comments
- Track modifications

Notifications

System must notify users when:

- Applications are submitted
- Approvals are required
- Applications are approved/rejected
- Appraisal cycles open/close

Dashboard

Role-based dashboards:

- Employee dashboard
- Manager dashboard
- HR dashboard
- Executive dashboard

Non-Functional Business Requirements (Non-Technical)

- Policy enforcement must be automated.
- Approval authority must be role-based.
- Data must be confidential and access-controlled.
- Reports must be exportable.
- The system must support organizational growth.

Success Metrics

- Reduction in manual HR processing time
- Reduction in approval delays
- Improved policy compliance
- Improved performance visibility
- Reduced financial risk from loans

Phase Scope

This PRD covers:

- Leave Management
- Staff Loan Management
- Staff Appraisal Management

Excludes:

- Payroll module
- Recruitment
- Attendance tracking

- Multi-company tenancy

Final Assessment

This is a rule-driven internal enterprise system with:

- Multi-stage workflow management
- Policy validation engine
- Weighted scoring mechanism
- Financial tracking logic
- Performance analytics layer

HRM Application – Phase 2 (Technical Requirements)

Product: HRM Application

Architecture Model: Containerized Microservice-Oriented Deployment

Target Environment: VPS (Production)

Stack: Next.js + Django REST Framework + PostgreSQL + Redis + Celery

System Architecture Overview

The HRM Application shall be implemented using a distributed containerized architecture composed of:

- Frontend Client (Next.js)
- Backend API (Django REST Framework)
- Relational Database (PostgreSQL)
- Message Broker & Cache (Redis)
- Background Task Processor (Celery Worker + Beat)
- Reverse Proxy & SSL Termination (Traefik)

- CI/CD Pipeline (GitHub Actions)
- VPS Infrastructure (Linux-based)

Architectural Pattern

- RESTful API architecture
 - Stateless backend services
 - Asynchronous task processing
 - Container isolation per service
 - Internal Docker network segmentation
 - HTTPS enforced via reverse proxy
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Technology Stack Requirements

Frontend Layer

Framework: Next.js (App Router or Pages Router)

Runtime: Node.js 20+

Rendering Strategy:

- CSR for authenticated dashboard views
- SSR for public-facing content (if applicable)

Functional Responsibilities:

- Authentication interface
- Role-based dashboards (Admin, HR, Manager, Staff)
- Leave submission & approval UI
- Payroll viewing interface
- Employee directory

- Notifications UI
- File upload interface

Security Requirements:

- JWT-based authentication storage (HTTP-only cookies preferred)
 - Environment-based API endpoint configuration
 - Strict CORS enforcement
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Backend Layer

Framework: Django 4.x

API Layer: Django REST Framework

Server: Gunicorn with Uvicorn workers

Core Capabilities

- JWT authentication (SimpleJWT)
- Role-Based Access Control (RBAC)
- Leave management workflow engine
- Payroll computation engine
- Approval chains
- Audit logging
- File handling
- Reporting endpoints

Performance Requirements

- Minimum 3 worker processes
- Connection pooling enabled
- Query optimization with indexed foreign keys

- Response time target: < 300ms for standard queries

Required Python Dependencies

- django-restframework
 - djangorestframework-simplejwt
 - psycopg2-binary
 - django-cors-headers
 - django-filter
 - django-redis
 - celery
 - redis
 - gunicorn
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Database Layer

Engine: PostgreSQL 15+

Requirements

- UTF-8 encoding
- WAL logging enabled
- Automated daily backups
- Indexed foreign keys
- Composite indexes for:
 - (employee_id, status)
 - (department_id, role)
- Data retention compliance
- Soft-delete strategy for HR records

Data Integrity Requirements

- Enforced foreign key constraints
- Transaction atomicity for payroll generation
- Audit trail tables for critical HR events

Cache & Message Broker

Engine: Redis

Usage Scope

- Celery broker
- Celery result backend
- Dashboard caching
- Rate limiting
- Session optimization (optional)

Constraints

- Single Redis instance acceptable for MVP
- Protected within internal Docker network
- No external exposure

Background Processing

Framework: Celery

Components:

- celery-worker
- celery-beat

Tasks to Offload

- Email notifications

- Leave status alerts
- Payroll monthly processing
- Scheduled report generation
- System reminders

Non-Functional Requirement

- All long-running tasks must be asynchronous.
- API endpoints must not block on external service calls.

Containerization

All services must be deployed using Docker.

Each service must run in an isolated container:

- frontend
- backend
- postgres
- redis
- celery-worker
- celery-beat
- traefik

Network Requirements

- Single internal Docker bridge network
- Only Traefik exposed publicly
- No direct host exposure for DB or Redis

Reverse Proxy & SSL

Proxy: Traefik

Responsibilities:

- Automatic Let's Encrypt SSL provisioning
 - Certificate auto-renewal
 - Domain routing:
 - app.domain.com → Frontend
 - api.domain.com → Backend
 - Security headers
 - Rate limiting middleware
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CI/CD Requirements

Platform: GitHub Actions

Trigger Conditions

- Push to main branch
- Pull request to main (test-only pipeline)

Required Pipeline Steps

1. Run backend tests
2. Run frontend tests
3. Build Docker images
4. Push images to container registry
5. SSH into VPS
6. Pull updated images
7. Restart services using docker compose

Deployment Constraint

- Zero manual production deployment steps.
 - Deployment must be fully automated.
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Security Requirements

Application-Level

- HTTPS enforced
- Secure cookie flags enabled
- CSRF protection enabled
- CORS restricted to production domain
- Rate limiting enabled
- Password hashing (PBKDF2 / Argon2)

Infrastructure-Level

- UFW firewall enabled
 - SSH key-based login only
 - Root login disabled
 - Fail2ban configured
 - Regular OS patching schedule
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Backup & Disaster Recovery

Required

- Daily PostgreSQL dump (automated cron)

- Weekly full VPS snapshot
- Backup retention: 30 days
- Off-site backup storage (S3 recommended)

Recovery Objective Targets

- RPO (Recovery Point Objective): ≤ 24 hours
 - RTO (Recovery Time Objective): ≤ 4 hours
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Monitoring & Observability

Minimum required:

- Application logging (structured logs)
- Celery task monitoring
- Traefik access logs
- Docker health checks

Recommended:

- Prometheus
 - Grafana
 - Sentry for error monitoring
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Scalability Requirements (Phase 2)

The architecture must support:

- Horizontal scaling of backend containers
- Migration to managed PostgreSQL

- Migration to managed Redis
 - CDN integration for static/media
 - Load balancing across multiple nodes
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Performance Targets

Metric	Target
API Response Time	< 300ms
Concurrent Users	50–100
Email Dispatch Latency	< 30 seconds
Payroll Batch Execution	< 2 minutes

Compliance & Data Protection

- Employee PII encryption at rest
 - Role-based data isolation
 - Audit log immutability
 - Access logging
 - GDPR-aligned data deletion capability (if applicable)
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Phase Exit Criteria

The Technical Requirements Phase shall be considered complete when:

- Infrastructure blueprint is approved
- Docker architecture defined

- CI/CD pipeline documented
- Security baseline validated
- Backup & recovery strategy validated
- Performance targets defined

HRM Application – Phase 3 (Database + API Blueprint)

Project Overview

Goal: Build a multi-tenant HRM system with employee management, leave workflow, attendance, payroll, and audit logging.

Modules:

1. Authentication & Access Control
 2. Organization Structure
 3. Employee Management
 4. Leave Management
 5. Attendance Management
 6. Payroll Management
 7. System Audit
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Database Schema

Organization Structure

Table	Fields	Notes
organizations	id (PK), name, registration_number, created_at, updated_at	Multi-tenant scope

departments	id (PK), organization_id (FK), name, description, created_at	1 org → N departments
positions	id (PK), department_id (FK), title, level, created_at	1 dept → N positions

Users & Roles

Table	Fields	Notes
users	id (PK), organization_id (FK), email, password_hash, status, last_login, created_at	Employee login account
roles	id (PK), name, description	Role definitions
user_roles	id (PK), user_id (FK), role_id (FK)	M:N user-role mapping

Employees

Table	Fields	Notes
employees	id (PK), user_id (FK nullable), employee_code, first_name, last_name, gender, dob, hire_date, employment_type, department_id (FK), position_id (FK), manager_id (FK self), status, created_at	Core employee data
employee_documents	id (PK), employee_id (FK), document_type, file_url, uploaded_at	Store on S3

Leave Management

Table	Fields	Notes
leave_types	id (PK), organization_id (FK), name, description, default_days, requires_approval, created_at	Annual, Sick, Casual, Maternity
leave_policies	id (PK), leave_type_id (FK), department_id (FK nullable), position_level, annual_entitlement, carry_forward_limit, created_at	Role/Dept-based policies

leave_balances	id (PK), employee_id (FK), leave_type_id (FK), year, allocated_days, used_days, carried_forward_days, remaining_days	Pre-calculated per year
leave_requests	id (PK), employee_id (FK), leave_type_id (FK), start_date, end_date, total_days, reason, status, approved_by (FK), approved_at, created_at	Workflow entity
leave_request_logs	id (PK), leave_request_id (FK), action, action_by (FK), timestamp, comment	Audit trail

Attendance

Table	Fields	Notes
attendance_records	id (PK), employee_id (FK), date, check_in, check_out, status, created_at	Manual or automated check-in/out

Payroll

Table	Fields	Notes
salary_structures	id (PK), employee_id (FK), basic_salary, allowance, deductions, effective_from	Active salary info
payroll_records	id (PK), employee_id (FK), month, year, gross_salary, total_deductions, net_salary, status, paid_at	Generated monthly

Audit Logs

Table	Fields	Notes
activity_logs	id (PK), user_id (FK), action, ip_address, created_at	Tracks all user actions

API Endpoints & Mapping

All endpoints follow [/api/v1](#) prefix.

A. Auth & Access Control

Endpoint	Method	Purpose	Roles
/auth/login	POST	Login	All
/auth/logout	POST	Logout	All
/auth/me	GET	Current user info	All

Endpoint	Methods	Description	Access
/users	GET, POST	List/Create users	HR/Admin
/users/:id	GET, PATCH, DELETE	Retrieve/Update/Delete user	HR/Admin
/users/:id/roles	POST, DELETE	Assign/Remove role	Admin
/roles	GET, POST	List/Create roles	Admin
/roles/:id	PATCH, DELETE	Update/Delete role	Admin

Organization Structure

Endpoint	Method	Purpose	Roles
/departments	GET, POST	List/Create	HR/Admin
/departments/:id	GET, PATCH, DELETE	CRUD	HR/Admin

/positions	GET, POST	List/Create	HR/Admin
/positions/:id	PATCH, DELETE	CRUD	HR/Admin

Employee Management

Endpoint	Method	Purpose	Roles
/employees	GET, POST	List/Create	HR/Admin
/employees/:id	GET, PATCH, DELETE	Retrieve/Update/Delete	HR/Admin
/employees/:id/documents	GET, POST	Document upload/list	HR/Admin
/documents/:id	DELETE	Delete document	HR/Admin
/employees/:id/subordinates	GET	Reports to employee	Manager/HR/Admin

Leave Management

Endpoint	Method	Purpose	Roles
/leave-types	GET, POST, PATCH, DELETE	Manage leave types	HR/Admin
/leave-policies	GET, POST, PATCH, DELETE	Manage leave rules	HR/Admin
/leave-balances	GET	View balances	Manager/HR/Admin
/employees/:id/leave-balances	GET	Employee-specific	Employee/Manager/HR
/leave-requests	GET, POST	List/Submit leave	Employee/Manager/HR
/leave-requests/:id	GET, PATCH	Retrieve/Update	Manager/HR
/leave-requests/:id/approve	POST	Approve leave	Manager/HR

/leave-requests/:id/reject	POST	Reject leave	Manager/HR
/leave-requests/:id/cancel	POST	Cancel leave	Employee
/leave-requests/:id/logs	GET	Audit logs	Manager/HR/Admin

Attendance

Endpoint	Method	Purpose	Roles
/attendance	GET	List attendance	Manager/HR/Admin
/attendance/check-in	POST	Employee check-in	Employee
/attendance/check-out	POST	Employee check-out	Employee
/attendance/manual	POST	HR override	HR/Admin
/employees/:id/attendance	GET	Employee attendance history	Employee/Manager/HR

Payroll

Endpoint	Method	Purpose	Roles
/salary-structures	GET, POST, PATCH	Manage salaries	HR/Admin
/payroll/generate	POST	Generate payroll	HR/Admin
/payroll	GET	List payroll	HR/Admin
/payroll/:id/pay	POST	Mark paid	HR/Admin

Audit & Logs

Endpoint	Method	Purpose	Roles
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/activity-logs	GET	View audit logs	Admin
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Phase Mapping – MVP vs Later

Phase	Modules	Tables	API Endpoints
Phase 1 (MVP)	Auth, Users/Roles, Employees, Departments, Positions, Leave Requests/Types	organizations, departments, positions, users, roles, user_roles, employees, leave_types, leave_requests	/auth/*, /users/*, /roles/*, /employees/*, /departments/*, /positions/*, /leave-types/*, /leave-requests/*
Phase 2	Leave Policies, Leave Balances, Employee Documents, Attendance	leave_policies, leave_balances, employee_documents, attendance_records	/leave-policies/*, /leave-balances/*, /employees/:id/documents, /attendance/*
Phase 3	Payroll & Audit	salary_structures, payroll_records, activity_logs	/salary-structures/*, /payroll/*, /activity-logs